

COMMON LOCALISATION COMBINATIONS

Finding clusters that localise together

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Teaching rule: one sign suggests a structure, two signs localise a lesion, and three concordant signs often make the localisation hard to miss.

Cortex & deep hemispheric patterns

Findings that occur together	Think of / localisation	Why the combination works
Aphasia + right face/arm weakness	Left dominant MCA cortex	Language cortex and lateral motor cortex lie close together.
Neglect + left face/arm weakness	Right nondominant parietal/MCA cortex	Right parietal association cortex and lateral motor cortex are involved together.
Leg weakness > arm + abulia + urinary incontinence	ACA / medial frontal lobe	Leg motor cortex and frontal initiation/micturition networks are medial.
Dense face-arm-leg weakness + no aphasia/neglect	Internal capsule	Compact motor fibres are affected without cortical involvement.
Face-arm-leg numbness + little/no weakness	Thalamus	Sensory relay nuclei are clustered together.
Hemiplegia + hemisensory loss + hemianopia	Anterior choroidal / internal capsular region	Motor, sensory and visual projection fibres converge deeply.
Homonymous hemianopia + otherwise normal neurological exam	Occipital cortex	A pure posterior visual pathway lesion can spare motor, sensory and language functions.
Bitemporal hemianopia + endocrine symptoms	Optic chiasm / sellar region	Crossing nasal retinal fibres lie above the pituitary region.
Alexia + right hemianopia + preserved writing	Left occipital cortex + splenium	Visual information cannot reach the dominant language cortex.
Acalculia + agraphia + finger agnosia + right-left confusion	Dominant angular gyrus	Classic inferior parietal cluster.

Brainstem combinations

Findings that occur together	Think of / localisation	Why the combination works
Ipsilateral cranial nerve palsy + contralateral limb weakness	Brainstem	Cranial-nerve fascicles are ipsilateral; corticospinal tracts produce opposite body signs.
III palsy + opposite hemiparesis	Midbrain — Weber pattern	CN III fascicle plus cerebral peduncle involvement.
III palsy + opposite tremor/ataxia	Midbrain tegmentum	CN III plus red nucleus / superior cerebellar pathway involvement.
VI palsy + opposite hemiparesis	Medial pons	VI fascicle and corticospinal tract run nearby.
LMN VII palsy + opposite hemiparesis	Pontine lesion	Facial fascicle with corticospinal tract.
Ipsilateral facial pain-temperature loss + contralateral body pain-temperature loss	Lateral brainstem	Trigeminal sensory pathway and spinothalamic tract are affected together.
Above + dysphagia/hoarseness + Horner + ataxia	Lateral medulla — Wallenberg	Nucleus ambiguus, sympathetic pathway and cerebellar connections are involved.
Facial weakness + deafness + vertigo + ipsilateral ataxia	Lateral pons / AICA	VII, VIII and cerebellar pathways lie together.
Tongue wasting/deviation + opposite hemiparesis	Medial medulla	XII fascicle plus pyramid.
Horizontal gaze palsy + LMN facial palsy	Facial colliculus / dorsal pons	VI nucleus/PPRF lies next to the looping VII fascicle.
Gaze palsy + INO	One-and-a-half syndrome	PPRF/VI nucleus plus ipsilateral MLF.
One-and-a-half + ipsilateral facial palsy	Eight-and-a-half syndrome	Adds nearby VII fascicle involvement.

Ocular & parasellar combinations

Findings that occur together	Think of / localisation	Why the combination works
Ptosis + down-and-out eye + dilated pupil	Compressive CN III palsy	Superficial parasympathetic fibres are vulnerable to compression.
Ptosis + down-and-out eye + pupil spared	Microvascular CN III palsy	Somatic fibres affected more than superficial pupillary fibres.
Vertical diplopia worse downstairs + compensatory head tilt	CN IV palsy	Typical superior oblique weakness pattern.
Horizontal diplopia + one eye cannot abduct	CN VI palsy	Lateral rectus / VI pathway lesion.
Adduction failure + opposite abducting nystagmus	MLF / INO	Disconnects conjugate horizontal gaze.
III + IV + VI palsies + V1/V2 numbness	Cavernous sinus	Multiple ocular motor nerves and trigeminal divisions run together.
Ophthalmoplegia + V1 loss + visual loss/RAPD	Orbital apex	Adds optic nerve involvement to ocular motor dysfunction.
VI palsy + Horner syndrome	Cavernous sinus / parasellar carotid region	VI nerve and sympathetic plexus lie close to the internal carotid.
Upgaze palsy + convergence-retraction nystagmus + light-near dissociation	Dorsal midbrain / Parinaud region	Classic tectal/pretectal combination.

Spinal cord combinations

Findings that occur together	Think of / localisation	Why the combination works
LMN arms + UMN legs	Cervical spinal cord	Segmental arm LMN signs with long-tract signs below.
UMN legs + normal arms + sensory level	Thoracic spinal cord	Leg corticospinal tracts involved below cervical enlargement.
Bilateral weakness + sensory level + bladder dysfunction	Spinal cord	Classic cord syndrome.
Ipsilateral weakness + ipsilateral vibration loss + opposite pain-temperature loss	Hemicord / Brown-Séquard	Different long tracts cross at different levels.
Motor paralysis + pain-temperature loss + preserved vibration	Anterior cord	Anterior spinal cord syndrome spares posterior columns.
Loss of vibration/proprioception + positive Romberg + preserved pain-temperature	Posterior columns	Dorsal column dysfunction.
Cape-like pain-temperature loss + preserved vibration + hand wasting	Central cervical cord / syringomyelia	Central cord lesion with segmental anterior horn involvement.
Segmental LMN signs at level + UMN signs below	Cord lesion at that segment	Combines local segmental damage with long-tract involvement.
Back/root pain + asymmetric Brown-Séquard-type signs	Extramedullary cord compression	Compression often begins eccentrically with root irritation.
Dissociated sensory loss + sacral sparing + segmental LMN signs	Intramedullary cord lesion	Characteristic intramedullary pattern.

Conus–cauda–epiconus patterns

Findings that occur together	Think of / localisation	Why the combination works
Early urinary retention + symmetric saddle anesthesia	Conus medullaris	Sacral segments densely packed centrally.
Saddle anesthesia + bowel/sexual dysfunction + little radicular pain	Conus	Sphincters and perineal sensation predominate.
Severe radicular pain + asymmetric flaccid weakness	Cauda equina	Multiple painful root lesions produce LMN asymmetry.
Absent ankle/knee reflexes + patchy root sensory loss + flexor plantars	Cauda equina	Root pattern without corticospinal signs.
Distal LMN leg signs + occasional Babinski + less abrupt sphincter symptoms	Epiconus	Mixed UMN-LMN pattern from lesion just above the conus.

Root–nerve–plexus combinations

Findings that occur together	Think of / localisation	Why the combination works
Neck/back pain + dermatomal sensory loss + myotomal weakness + reflex loss	Root	The root gives a dermatome, myotome and reflex arc.
Weakness involving different named nerves but sharing one myotome	Root	Points proximal to the named peripheral nerves.
Motor + sensory deficit fitting one named nerve	Peripheral nerve	Single nerve territory involvement.
Multiple named nerves involved in one limb + patchy sensory loss	Plexus	Pattern does not fit one root or one nerve.
Foot drop + eversion weak + inversion preserved	Common peroneal nerve	Tibialis posterior (inversion) is spared.
Foot drop + inversion weak + hip abduction weak	L5 root	Adds non-peroneal L5 muscles.
Foot drop + plantarflexion also weak + ankle jerk absent	Sciatic nerve	Both tibial and peroneal divisions are affected.
Wrist drop + triceps preserved	Radial nerve at spiral groove	Lesion is distal to triceps branches.
Finger drop + wrist extension preserved + no sensory loss	Posterior interosseous nerve	Pure motor branch of radial nerve.
Cannot form 'OK' sign + no sensory loss	Anterior interosseous nerve	Pure motor median branch.

Ataxia combinations

Findings that occur together	Think of / localisation	Why the combination works
Dysmetria + intention tremor + dysdiadochokinesia on same side	Ipsilateral cerebellar hemisphere	Classic cerebellar limb signs.
Broad-based gait + truncal instability + relatively normal arms	Cerebellar vermis	Predominantly axial involvement.
Ataxia + loss of vibration/proprioception + positive Romberg	Sensory pathway / posterior columns	Proprioceptive loss causes sensory ataxia.
Ataxia worse in darkness	Sensory ataxia	Visual compensation is lost.
Ataxia already severe with eyes open	Cerebellar ataxia	Not mainly dependent on visual compensation.
Vertigo + nystagmus + gait veering	Vestibular system	Vestibular imbalance causes directional falling.
Vertigo + limb dysmetria/diplopia/dysarthria	Central — brainstem/cerebellum	Additional central signs localise posterior circulation pathways.

Bulbar & lower cranial nerve combinations

Findings that occur together	Think of / localisation	Why the combination works
Hoarseness + palatal droop + weak cough	CN X / nucleus ambiguus	Shared vagal/bulbar pathway involvement.
Palatal weakness + absent gag + shoulder weakness	IX–X–XI / jugular foramen	Lower cranial nerves cluster here.
IX–X–XI palsies + XII palsy	Collet-Sicard region	Adds hypoglossal involvement near skull base.
IX–X–XI–XII palsies + Horner syndrome	Villaret / retroparotid space	Adds sympathetic chain involvement.
Tongue wasting + fasciculation + absent/reduced gag	Bulbar LMN system	Lower motor neuron bulbar pattern.
Dysarthria/dysphagia + brisk jaw jerk + emotional lability	Bilateral corticobulbar / pseudobulbar palsy	Upper motor neuron bulbar pattern.
Vagus + hypoglossal palsy with other lower CN spared	Extracranial X/XII — Tapia pattern	Characteristic paired extracranial palsy.

Motor-system combinations

Findings that occur together	Think of / localisation	Why the combination works
Proximal symmetric weakness + normal sensation	Muscle	Myopathy usually causes symmetric proximal weakness without sensory loss.
Proximal weakness + ptosis/ophthalmoplegia + normal sensation/reflexes	Myasthenia / NMJ	Fatigable ocular-bulbar involvement suggests NMJ disease.
Proximal weakness + autonomic symptoms + reduced reflexes that facilitate	Lambert-Eaton	Presynaptic NMJ disorder.
Distal symmetric weakness + stocking sensory loss + absent reflexes	Peripheral neuropathy	Length-dependent neuropathic pattern.
Wasting + fasciculations + reduced reflexes + normal sensation	Anterior horn cell / LMN	Pure lower motor neuron system involvement.
Wasting/fasciculations + brisk reflexes/Babinski + normal sensation	Motor neuron disease	Combined UMN and LMN pattern.
Ascending symmetric weakness + areflexia	Guillain-Barré / polyradiculoneuropathy	Acute generalized LMN-root/nerve pattern.

The 12 must-know combinations

Findings that occur together	Think of / localisation	Why the combination works
Aphasia + right weakness	Left cerebral cortex	High-yield dominant hemispheric pattern.
Neglect + left weakness	Right parietal cortex	High-yield nondominant hemispheric pattern.
Dense pure hemiparesis without cortical signs	Internal capsule	Classic deep lesion pattern.
Ipsilateral CN + contralateral body signs	Brainstem	Core crossed finding.
Face one side + body pain loss opposite	Lateral brainstem	Crossed sensory pattern.
LMN arms + UMN legs	Cervical cord	Core cervical cord pattern.
UMN legs + sensory level	Spinal cord	Core cord pattern.
Ipsilateral motor/proprioception loss + contralateral pain loss	Hemicord	Brown-Séquard pattern.
Early bladder + symmetric saddle anesthesia	Conus	Core conus pattern.
Painful asymmetric areflexic legs + saddle anesthesia	Cauda equina	Core cauda pattern.
Dermatome + myotome + reflex	Root	Core radiculopathy pattern.
Named motor + sensory territory	Peripheral nerve	Core mononeuropathy pattern.